EXPERIMENT 3B: MODEL PLANNING AND BUILDING – REGRESSION

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Learning Objective

To build and evaluate regression models using a real-world dataset.

Problem Statement

A retail company wants to predict sales based on TV, Radio, and Newspaper advertising expenditure to optimize its advertising budget.

Tasks to Perform

1. Load the Advertising dataset from a URL.
2. Explore the dataset and check missing values, duplicates, and statistics.
3. Select TV, Radio, and Newspaper as input features and Sales as the target.
4. Generate a pair plot and correlation heatmap.
5. Identify the feature most correlated with Sales.
6. Split the dataset into 80% training and 20% testing data.
7. Build Linear Regression, Polynomial Regression, and SVR models.
8. Evaluate models using MAE, MSE, RMSE, and R^2 score.
9. Compare model performance using a bar chart.
10. Plot actual versus predicted Sales for the best model.
11. Identify the best model and suggest three advertising budget recommendations.

AIM

To build and compare Linear Regression, Polynomial Regression, and Support Vector Regression models to predict sales based on advertising expenditure.

ALGORITHM

1. Load the Advertising dataset from a URL and explore its features.
2. Select TV, Radio, and Newspaper as inputs and Sales as the target.
3. Perform EDA using a pair plot and correlation heatmap.
4. Split the dataset into 80% training and 20% testing data.
5. Train Linear Regression, Polynomial Regression, and SVR models.
6. Evaluate the models using MAE, MSE, RMSE, and R^2 score.
7. Compare model performance and plot actual versus predicted sales.

8. Identify the best model and suggest three advertising budget recommendations.

PROGRAM AND OUTPUT


```
# Question 1: Download and load the Advertising dataset from a public CSV URL.
# LUQMAN T (241801141)
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

url = "advertising.csv"
df = pd.read_csv(url)
# Remove the optional index column if present.
df = df.drop(columns=["Unnamed: 0"], errors="ignore")
print("Dataset loaded:", df.shape)
display(df.head())
```

```
Dataset loaded: (200, 4)
```

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	12.0
3	151.5	41.3	58.5	16.5
4	180.8	10.8	58.4	17.9

```
# Question 2: Explore the dataset and select TV, Radio, Newspaper as X and Sales as y.
# LUQMAN T (241801141)
display(df.head())
display(df.tail())
print("Shape:", df.shape)
print("\nData types:\n", df.dtypes)
print("\nMissing values:\n", df.isnull().sum())
print("Duplicates:", df.duplicated().sum())
display(df.describe())

X = df[["TV", "Radio", "Newspaper"]]
y = df["Sales"]
```

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	12.0
3	151.5	41.3	58.5	16.5
4	180.8	10.8	58.4	17.9

	TV	Radio	Newspaper	Sales
195	38.2	3.7	13.8	7.6
196	94.2	4.9	8.1	14.0
197	177.0	9.3	6.4	14.8
198	283.6	42.0	66.2	25.5
199	232.1	8.6	8.7	18.4

```
Shape: (200, 4)

Data types:
 TV           float64
Radio        float64
Newspaper    float64
Sales        float64
dtype: object

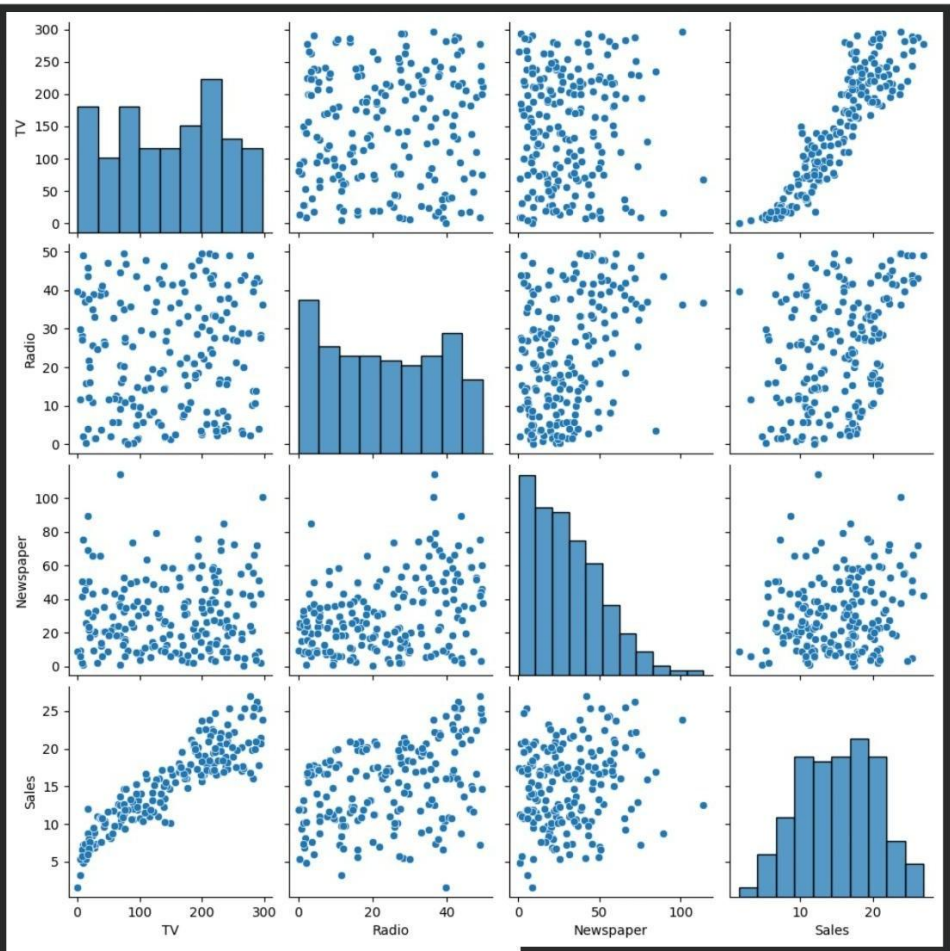
Missing values:
 TV           0
Radio        0
Newspaper    0
Sales        0
dtype: int64
Duplicates: 0
```

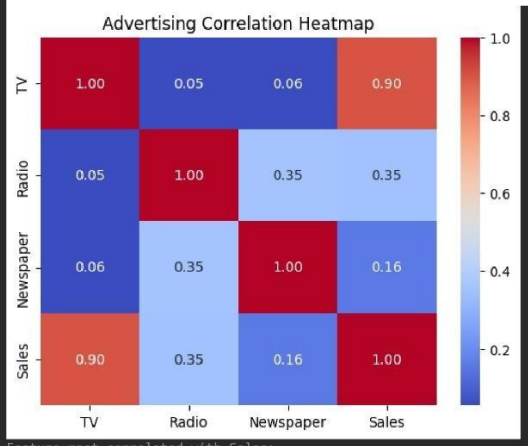
	TV	Radio	Newspaper	Sales
count	200.000000	200.000000	200.000000	200.000000
mean	147.042500	23.264000	30.554000	15.130500
std	85.854236	14.846809	21.778621	5.283892
min	0.700000	0.000000	0.300000	1.600000
25%	74.375000	9.975000	12.750000	11.000000
50%	149.750000	22.900000	25.750000	16.000000
75%	218.825000	36.525000	45.100000	19.050000
max	296.400000	49.600000	114.000000	27.000000

```
# Question 3: Generate a pair plot and correlation heatmap, and identify the feature most correlated with Sales.
# LUQMAN T (241801141)
sns.pairplot(df[["TV", "Radio", "Newspaper", "Sales"]])
plt.show()

corr = df[["TV", "Radio", "Newspaper", "Sales"]].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Advertising Correlation Heatmap")
plt.show()

print("Feature most correlated with Sales:")
print(corr["Sales"].drop("Sales").abs().sort_values(ascending=False).index[0])
```






```
Feature most correlated with Sales:
TV
```

```
# Question 4: Split the dataset into 80% training and 20% testing data.
# LUQMAN T (241801141)
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.pipeline import make_pipeline
from sklearn.linear_model import LinearRegression
from sklearn.svm import SVR

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)
print("Training set:", X_train.shape)
print("Testing set:", X_test.shape)
```

```
Training set: (160, 3)
Testing set: (40, 3)
```

```
# Question 5: Build Linear Regression, Polynomial Regression (degree 2), and SVR (RBF) models and predict Sales.
# LUQMAN T (241801141)
models = {
    "Linear Regression": LinearRegression(),
    "Polynomial Regression": make_pipeline(PolynomialFeatures(degree=2, include_bias=False), LinearRegression()),
    "SVR (RBF)": make_pipeline(StandardScaler(), SVR(kernel="rbf"))
}

predictions = {}
for name, model in models.items():
    model.fit(X_train, y_train)
    predictions[name] = model.predict(X_test)
    print(name, "predictions:", np.round(predictions[name][:5], 2))
```

```
Linear Regression predictions: [17.03 20.41 23.72  9.27 21.68]
Polynomial Regression predictions: [17.56 21.96 22.08  7.35 23.24]
SVR (RBF) predictions: [17.28 21.15 20.54 10.43 20.17]
```

```
# Question 6: Evaluate all models using MAE, MSE, RMSE, and R2 score.
# LUQMAN T (241801141)
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

results = []
for name, pred in predictions.items():
    mae = mean_absolute_error(y_test, pred)
    mse = mean_squared_error(y_test, pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_test, pred)
    results.append({"Model": name, "MAE": mae, "MSE": mse, "RMSE": rmse, "R2 Score": r2})

results_df = pd.DataFrame(results)
display(results_df.round(3))
```

	Model	MAE	MSE	RMSE	R2 Score
0	Linear Regression	1.275	2.908	1.705	0.906
1	Polynomial Regression	0.903	1.443	1.201	0.953
2	SVR (RBF)	1.555	4.421	2.103	0.857

```
# Question 7: Compare all models using a grouped bar chart and plot Actual vs Predicted Sales for the best model.
# LUQMAN T (241801141)
metrics = results_df.set_index("Model")
metrics.plot(kind="bar", figsize=(11, 6), rot=0)
plt.title("Regression Model Performance Comparison")
plt.ylabel("Metric Value")
plt.tight_layout()
plt.show()

# Lowest RMSE is used to select the best-performing model.
best_name = results_df.loc[results_df["RMSE"].idxmin(), "Model"]
best_pred = predictions[best_name]
print("Best model based on lowest RMSE:", best_name)

plt.figure(figsize=(6, 5))
plt.scatter(y_test, best_pred)
plt.xlabel("Actual Sales")
plt.ylabel("Predicted Sales")
plt.title(f"Actual vs Predicted Sales — {best_name}")
plt.tight_layout()
plt.show()
```


RESULT

The three regression models were built, evaluated, and compared to identify the best model for sales prediction.